SECURING NETWORK PERIMETERS USING FIREWALLS, IDS/IPS, AND VPNS

PRIYATHARSHINI V

727723EUCD042

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kali@kali: ~/openvpn-ca
Session  Actions  Edit  View  Help
└─$ sudo snort -T -c /etc/snort/snort.lua
o")~   Snort++ 3.10.2.0
-------------------------------------------------------------------------------
Loading /etc/snort/snort.lua:
        host_cache
        host_tracker
        packets
        search_engine
        output
        trace
        ips
        so_proxy
        process
        network
        hosts
        daq
        decode
        active
        alerts
Finished /etc/snort/snort.lua:
Loading ips.rules:
Loading /etc/snort/rules/local.rules:
Finished /etc/snort/rules/local.rules:
Finished ips.rules:
-------------------------------------------------------------------------------
ips policies rule stats
               id  loaded  shared enabled    file
                0       1       0       1    /etc/snort/snort.lua
-------------------------------------------------------------------------------
rule counts
        total rules loaded: 1
                text rules: 1
            option chains: 1
            chain headers: 1
-------------------------------------------------------------------------------
port rule counts
            tcp     udp    icmp      ip
    any       0       0       1       0
  total       0       0       1       0
-------------------------------------------------------------------------------
afpacket DAQ configured to passive.

Snort successfully validated the configuration (with 0 warnings).
o")~   Snort exiting

┌──(kali㉿kali)-[~]
└─$ sudo snort -c /etc/snort/snort.lua -i eth0
o")~   Snort++ 3.10.2.0
-------------------------------------------------------------------------------
Loading /etc/snort/snort.lua:
        active
        alerts
        daq
        decode
        host_cache
        host_tracker
        hosts
        process
        search_engine
        so_proxy
        output
        ips
        trace
        packets
        network
Finished /etc/snort/snort.lua:
Loading ips.rules:
Loading /etc/snort/rules/local.rules:
Finished /etc/snort/rules/local.rules:
Finished ips.rules:
```

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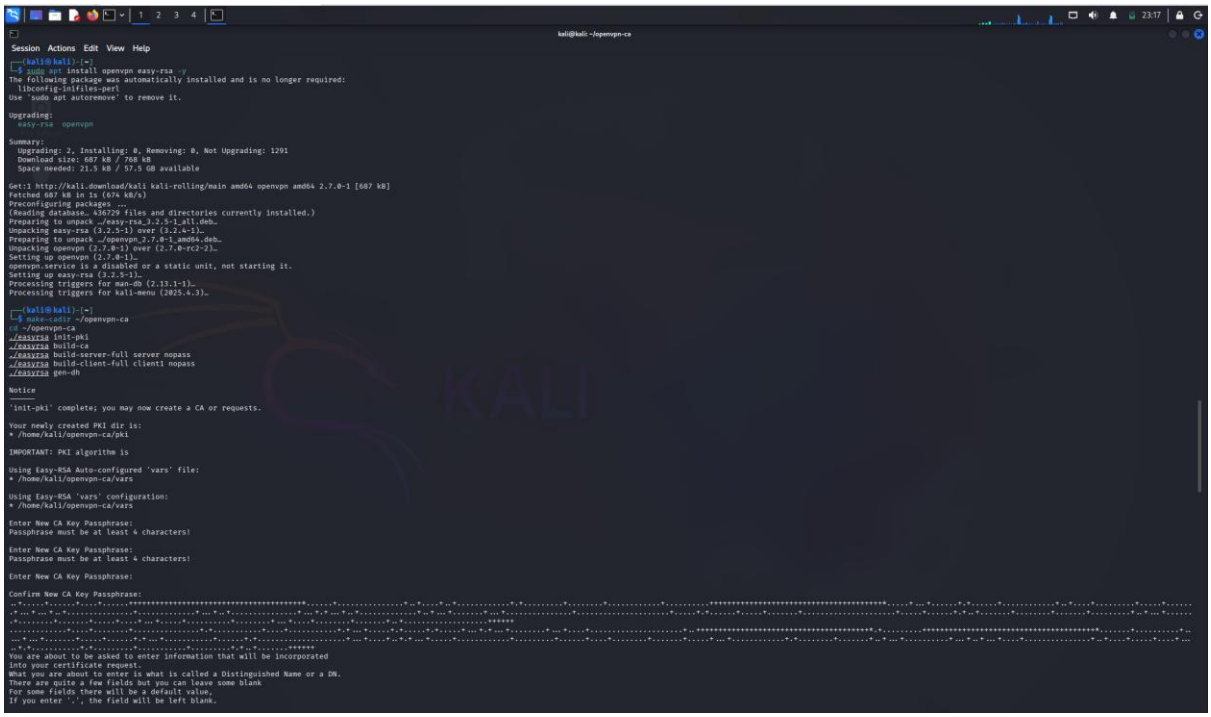
-------------------------------------------------------------------------------
ips policies rule stats
               id  loaded  shared enabled    file
                0       1       0       1    /etc/snort/snort.lua
-------------------------------------------------------------------------------
rule counts
        total rules loaded: 1
                text rules: 1
            option chains: 1
            chain headers: 1
-------------------------------------------------------------------------------
port rule counts
            tcp     udp    icmp      ip
    any       0       0       1       0
  total       0       0       1       0
-------------------------------------------------------------------------------
afpacket DAQ configured to passive.
Commencing packet processing
Retry queue interval is: 200 ms
++ [0] eth0
^C** caught int signal
== stopping
-- [0] eth0
-------------------------------------------------------------------------------
Packet Statistics
-------------------------------------------------------------------------------
daq
                 received: 7
                 analyzed: 6
                    allow: 6
                     idle: 30
                 rx_bytes: 554
-------------------------------------------------------------------------------
codec
                    total: 6          (100.000%)
                      eth: 6          (100.000%)
                    icmp6: 1          ( 16.667%)
                     ipv4: 3          ( 50.000%)
                     ipv6: 3          ( 50.000%)
                      udp: 5          ( 83.333%)
-------------------------------------------------------------------------------
Module Statistics
-------------------------------------------------------------------------------
detection
                 analyzed: 6
                hard_evals: 1
                   alerts: 1
             total_alerts: 1
                   logged: 1
-------------------------------------------------------------------------------
ips_actions
                    alert: 1
-------------------------------------------------------------------------------
search_engine
         qualified_events: 1
-------------------------------------------------------------------------------
stream
     no_flow_no_inspector: 6
-------------------------------------------------------------------------------
Summary Statistics
-------------------------------------------------------------------------------
process
                  signals: 1
-------------------------------------------------------------------------------
timing
                  runtime: 00:00:33
                  seconds: 33.503584
o")~   Snort exiting

┌──(kali㉿kali)-[~]
└─$ sudo apt install openvpn easy-rsa -y
The following package was automatically installed and is no longer required:
  libconfig-inifiles-perl
```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Session Actions Edit View Help
┌──(kali㉿kali)-[~]
└─$ sudo apt install openvpn easy-rsa -y
The following package was automatically installed and is no longer required:
libconfig-inifiles-perl
Use 'sudo apt autoremove' to remove it.
Upgrading:
easy-rsa openvpn
Summary:
Upgrading: 2, Installing: 0, Removing: 0, Not Upgrading: 1291
Download size: 687 kB / 768 kB
Space needed: 21.5 kB / 57.5 GB available
Get:1 http://kali.download/kali kali-rolling/main amd64 openvpn amd64 2.7.0-1 [687 kB]
Fetched 687 kB in 1s (674 kB/s)
Preconfiguring packages ...
(Reading database... 436729 files and directories currently installed.)
Preparing to unpack .../easy-rsa_3.2.5-1_all.deb...
Unpacking easy-rsa (3.2.5-1) over (3.2.4-1)...
Preparing to unpack .../openvpn_2.7.0-1_amd64.deb...
Unpacking openvpn (2.7.0-1) over (2.7.0-rc2-2)...
Setting up openvpn (2.7.0-1)...
openvpn.service is a disabled or a static unit, not starting it.
Setting up easy-rsa (3.2.5-1)...
Processing triggers for man-db (2.13.1-1)...
Processing triggers for kali-menu (2025.4.3)...
┌──(kali㉿kali)-[~]
└─$ make-cadir ~/openvpn-ca
cd ~/openvpn-ca
./easyrsa init-pki
./easyrsa build-ca
./easyrsa build-server-full server nopass
./easyrsa build-client-full client1 nopass
./easyrsa gen-dh
Notice
'init-pki' complete; you may now create a CA or requests.
Your newly created PKI dir is:
* /home/kali/openvpn-ca/pki
IMPORTANT: PKI algorithm is
Using Easy-RSA Auto-configured 'vars' file:
* /home/kali/openvpn-ca/vars
Using Easy-RSA 'vars' configuration:
* /home/kali/openvpn-ca/vars
Enter New CA Key Passphrase:
Passphrase must be at least 4 characters!
Enter New CA Key Passphrase:
Passphrase must be at least 4 characters!
Enter New CA Key Passphrase:
Confirm New CA Key Passphrase:
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.

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Session Actions Edit View Help
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
Common Name (eg: your user, host, or server name) [Easy-RSA CA]:hello
Notice
CA creation complete. Your new CA certificate is at:
* /home/kali/openvpn-ca/pki/ca.crt
Build-ca completed successfully.
Using Easy-RSA 'vars' configuration:
* /home/kali/openvpn-ca/vars
Notice
Private-Key and Public-Certificate-Request files created.
Your files are:
* req: /home/kali/openvpn-ca/pki/reqs/server.req
* key: /home/kali/openvpn-ca/pki/private/server.key
You are about to sign the following certificate:
Requested CN: 'server'
Requested type: 'server'
Valid for: '825' days
subject=
commonName = server
Type the word 'yes' to continue, or any other input to abort.
Confirm requested details: hello
Notice
Aborting without confirmation.
Using Easy-RSA 'vars' configuration:
* /home/kali/openvpn-ca/vars
Notice
Private-Key and Public-Certificate-Request files created.
Your files are:
* req: /home/kali/openvpn-ca/pki/reqs/client1.req
* key: /home/kali/openvpn-ca/pki/private/client1.key
You are about to sign the following certificate:
Requested CN: 'client1'
Requested type: 'client'
Valid for: '825' days

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kali@kali: ~/openvpn-ca
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Your files are:
* req: /home/kali/openvpn-ca/pki/reqs/client1.req
* key: /home/kali/openvpn-ca/pki/private/client1.key

You are about to sign the following certificate:

  Requested CN:     'client1'
  Requested type:   'client'
  Valid for:        '825' days


subject=
    commonName                = client1

Type the word 'yes' to continue, or any other input to abort.
  Confirm requested details: hello


Notice
------
Aborting without confirmation.

Using Easy-RSA 'vars' configuration:
* /home/kali/openvpn-ca/vars
Generating DH parameters, 2048 bit long safe prime
.....+.....+.....+
```